

Optimising the Positron Angle-Energy Weighting for the Electric Dipole Moment Search at the Fermilab muon $g-2$ experiment

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Declaration

I declare that this dissertation is my own work and has not been submitted in the same or substantially similar form for the award of any other degree at this or any other institution. All sources of information have been properly acknowledged.

Generative AI tools were used in a limited capacity to assist with proofreading, clarification of concepts, and presentation. No content was generated or included without critical evaluation, and all analysis, derivations, text, and conclusions are my own, unless otherwise stated.

Abstract

Searches for a permanent electric dipole moment (EDM) of a fundamental particle, such as the muon, provide a sensitive probe of physics beyond the Standard Model (SM), as a non-zero EDM would indicate additional sources of $\mathbb{CP}$ violation. In the Fermilab muon $g-2$ experiment, a muon EDM would manifest as a small out-of-plane precession of the muon spin, producing a vertical asymmetry in the angular and energy distribution of decay positrons, characterised by a tilt δ of the spin precession plane – implying that the EDM information is not uniformly distributed across positron angle-energy phase space. In this project, the sensitivity to the muon EDM is maximised by constructing a weighting scheme using idealised datasets generated from Born-level analytic PDFs, which enhances regions with the greatest EDM sensitivity, quantified using a figure of merit (FOM).

As an initial approach, an optimal weighting scheme was obtained numerically using the Covariance Matrix Adaptation Evolution Strategy (CMA-ES), where the FOM was optimised for 32 distinct fractional energies λ and interpolated to obtain a full weight map. Applying this weight map gave a 49.3% increase in sensitivity to the muon EDM, corresponding to an effective increase in statistical power.

To understand and validate the structure found by the numerical optimiser, an analytic stationarity condition for the optimal weighting scheme was derived by treating the FOM as a functional of the weight. The analytic and numerical weights were found to be in close agreement, with a cosine similarity score of 0.996 (where 1.0 indicates perfect agreement). Applying the analytic weight gave a 49.1% improvement in the FOM, confirming that the numerical optimisation captures the same sensitivity structure as the analytic stationarity solution.

A further study extracted the EDM tilt parameter δ from the observables. In the unweighted case, the extracted value of $\delta = (80.0 \pm 0.1) \text{ mrad}$ is consistent with the injected value of $\delta = 80 \text{ mrad}$. Applying the optimal weighting makes the observable approximately linear in fractional lab-frame energy, simplifying the extraction of δ and improving the statistical precision relative to the likelihood-based approach. An independent log-likelihood extraction provides a consistent result with a larger uncertainty, confirming the robustness of the weighting-based method.

Although this study is based on idealised toy data derived from Born-level analytic PDFs and does not include experimental effects such as detector resolution or pileup, the framework is in principle transferable to real Fermilab $g-2$ datasets. It can also be readily adapted to future spin-based muon EDM experiments. These results demonstrate that optimised weighting of positron kinematics can significantly enhance sensitivity to a muon EDM and provide a useful complement to conventional likelihood-based methods in future precision measurements.



1 Introduction

Known fundamental particle production mechanisms generate matter and antimatter symmetrically, implying that the Universe should contain equal amounts of both. Observationally, however, the Universe is overwhelmingly matter dominated [1], indicating that processes must exist that favour the production of baryons over antibaryons.

The conditions required to generate such an asymmetry were first outlined by Andrei Sakharov [2] in 1967, and are discussed in **Section 3.5.3**. Among these is the requirement that both charge conjugation ($\mathbb{C}$) and charge-parity ($\mathbb{CP}$) symmetries are violated, as conservation of $\mathbb{CP}$ would ensure equal production rates for baryons and antibaryons.

Within the Standard Model (SM), $\mathbb{CP}$ violation arises through the complex phase of the Cabibbo–Kobayashi–Maskawa (CKM) matrix [3], but is insufficient to account for the observed matter–antimatter asymmetry, implying the presence of additional sources of $\mathbb{CP}$ violation.

A particularly sensitive probe of such effects is the existence of a permanent electric dipole moment (EDM) of a fundamental particle. Unlike magnetic dipole moments, which are allowed by all discrete symmetries (**Section 3.4**), a non-zero EDM violates time-reversal ($\mathbb{T}$) symmetry and, via the $\mathbb{CPT}$ theorem (**Section 3.5.1**), also implies $\mathbb{CP}$ violation. In the absence of such symmetry breaking, an EDM must vanish, making it a null test of $\mathbb{CP}$ symmetry.

Within the SM, EDMs arise only through higher-order weak processes, leading to extremely small predicted values far below current experimental sensitivity. Extensions of the SM, such as supersymmetric models and multi-Higgs doublet models, introduce additional $\mathbb{CP}$ -violating phases that can significantly enhance EDMs [4, 5]. The observation of a non-zero EDM would therefore provide clear evidence for physics beyond the Standard Model.

The Fermilab $g-2$ experiment provides the most precise determination of the muon¹ anomalous magnetic moment to date [6]. Muons are stored in a highly uniform laboratory-frame magnetic field and circulate at relativistic speeds ($\gamma \approx 29.3$), allowing precise measurement of the spin precession and hence the magnetic dipole moment.

In the muon rest frame, this motion generates an effective combination of electric and magnetic fields (see **App. D**), inducing spin precession about a well-defined axis. In the absence of an EDM, this precession occurs in a plane perpendicular to the magnetic field, whereas a non-zero EDM introduces a small out-of-plane component. The spin direction is inferred experimentally through the angular and energy distribution of the decay positrons, which are preferentially emitted along the spin direction.

Existing $g-2$ analyses employ energy-dependent event weighting to optimise the extraction of the anomalous precession frequency [6]. In the presence of an EDM, however, the sensitivity is encoded in both the angular and energy distributions of the decay positrons, suggesting that weighting over the full phase space may enhance sensitivity.

The approach taken in this project is outlined in the following section, where an optimal weighting scheme over positron angle–energy phase space is constructed to maximise sensitivity to a possible muon EDM.

¹For simplicity, “muon” is used throughout; the experiment operates with μ^+ , and statements apply to the antimuon unless otherwise specified.

2 Aims and Methodology

The aim of this project is to maximise the sensitivity of the Fermilab $g-2$ experiment to a potential muon electric dipole moment (EDM) by optimising the use of positron angle–energy information. This sensitivity depends on how information about the muon spin is distributed across the positron angle–energy phase space. Positrons emitted in specific regions provide the strongest correlation with the muon spin orientation, and therefore carry the greatest sensitivity to a potential EDM, as discussed in **Section 3.6**, and therefore the sensitivity is not uniformly distributed across phase space. Identifying which regions of positron decay angle and energy maximise sensitivity to an EDM signal is therefore a central objective.

To achieve this, a weighting scheme designed to maximise sensitivity is constructed which enhances the contribution from regions of phase space with the greatest sensitivity to the EDM signal, requiring a balance between signal strength and statistical uncertainty.

Two complementary approaches are employed to determine an optimised weighting. First, a numerical optimisation is performed to determine the optimal weighting. Secondly, motivated by the observation that the numerical optimiser converges to a well-defined structured solution, a self-consistent analytic stationary weight is derived. As a supplementary analysis, the injected EDM tilt parameter δ is independently extracted to validate the sensitivity of the weighting scheme, using both a fitting procedure and an independent log-likelihood approach.

3 Theoretical Background

3.1 The Standard Model

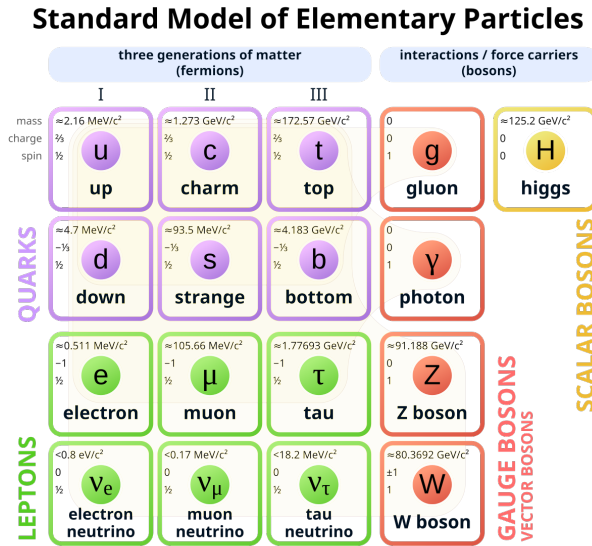


Figure 1: The Standard Model of particle physics, showing the three generations of fermions (quarks and leptons) and the bosons that mediate the fundamental interactions, along with the Higgs boson responsible for mass generation [7].

The Standard Model (SM) of particle physics (**Fig. 1**) provides a theoretical framework for describing all known fundamental particles and their interactions, with the exception of gravity [7, 8]. Particles are classified according to their intrinsic spin as either fermions (half-integer spin) or bosons (integer spin). Fermions, which constitute matter, are divided into quarks and leptons. Quarks carry colour charge and

therefore interact via the strong force, forming composite particles known as hadrons, whereas leptons do not and are fundamental. The gauge bosons mediate the electromagnetic, weak, and strong interactions, whilst the Higgs boson is responsible for mass generation through spontaneous symmetry breaking [9].

Although the SM has achieved remarkable experimental success, it remains incomplete. Not only does it not incorporate gravity, dark matter, or neutrino masses in a natural way, but it also does not contain sufficient charge-parity (CP) violation. One of the earliest experimental observations of CP violation was the 2π decay of K_2^0 mesons [10]. Within the SM, such effects arise through the complex phase of the Cabibbo–Kobayashi–Maskawa (CKM) [3] matrix, but are highly suppressed. This suppression arises from the structure of quark mixing, in which the weak and mass eigenstates are misaligned, resulting in small CP-violating effects in observable processes. However, the magnitude of CP violation generated in this way is insufficient to account for the observed matter–antimatter asymmetry of the Universe, falling many orders of magnitude below that required by cosmological observations. This discrepancy strongly suggests the presence of additional sources of CP violation beyond the Standard Model.

Extensions of the SM can introduce additional sources of CP violation through new interactions that give rise to electric dipole moments (EDMs) for fundamental particles. A non-zero EDM provides a particularly sensitive probe of such effects, as it directly signals CP violation and is highly suppressed within the Standard Model, making it a clean signature of new physics.

In this context, the muon is of particular interest. Experiments such as the Fermilab $g-2$ experiment are designed to measure the anomalous magnetic moment through the spin precession induced by the magnetic dipole moment. A non-zero EDM would introduce an additional contribution to this precession, providing a distinct signature that can be probed within the same experimental framework.

3.2 Production of Muons

In the Fermilab $g-2$ experiment [6], muons are produced by directing a high-intensity proton beam onto a dense Inconel target (a nickel alloy). The resulting proton–nucleus collisions produce a range of secondary mesons, of which the charged pions ($\pi^\pm$) are of primary importance, as they decay via the weak interaction into leptonic final states. The relevant decay channels are:

$$\pi^+ \rightarrow \mu^+ + \nu_\mu, \quad \pi^- \rightarrow \mu^- + \bar{\nu}_\mu. \quad (1)$$

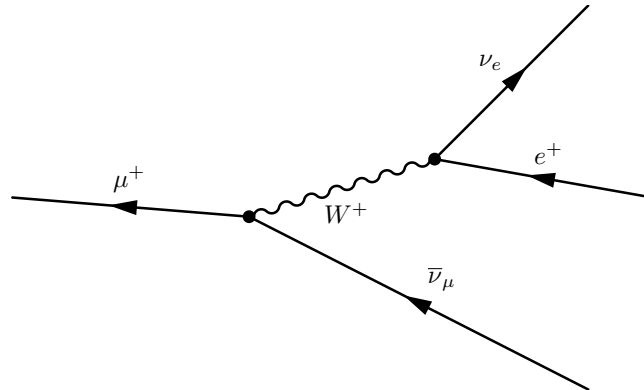


Figure 2: Born-level Feynman diagram for $\mu^+ \rightarrow e^+ + \nu_e + \bar{\nu}_\mu$ via W^+ exchange, illustrating the charged weak interaction responsible for muon decay.

The positively charged pions are magnetically focused and momentum-selected before decaying in flight, producing a collimated beam of highly polarised muons. These muons are subsequently injected into the storage ring, where their spin precession is measured with high precision to extract the anomalous

magnetic moment and to search for a possible EDM, as described in **Section 3.6**.

The subsequent decay of the muon proceeds through the charged weak current, shown schematically in **Fig. 2**, and is given by:

$$\mu^+ \rightarrow e^+ + \nu_e + \bar{\nu}_\mu. \quad (2)$$

As described in **Section 3.3**, the chiral structure of the weak interaction leads to a strong correlation between the muon spin direction and the direction of the emitted positron.

3.3 Helicity and Chirality

An important property of particles is their handedness, or helicity, denoted by $\mathfrak{h}$ and illustrated in **Fig. 3**. It is defined as the projection of a particle's spin $\mathbf{S}$ onto its momentum $\mathbf{p}$,

$$\mathfrak{h} = \frac{\mathbf{S} \cdot \mathbf{p}}{|\mathbf{p}|}, \quad (3)$$

where $|\mathbf{p}| = \sqrt{\mathbf{p} \cdot \mathbf{p}}$. For a particle with a finite, non-zero mass m , a Lorentz boost (see **App. C**) along its direction of motion to a frame moving faster than the particle corresponds to a transformation in which the particle is observed to move in the opposite direction (its 3-momentum changes sign), whilst the spin direction remains unchanged. Consequently, the helicity changes sign and is not invariant under Lorentz transformations. To describe an intrinsic notion of handedness that is independent of reference frame, the related quantity chirality is introduced, which is Lorentz invariant. In the Dirac formalism, chirality corresponds to the eigenstates of the γ^5 operator, distinguishing left- and right-handed components of a fermion field [8].

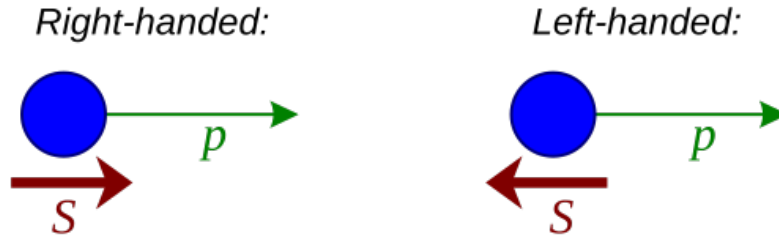


Figure 3: The two helicity states, $\mathfrak{h}$, for a spin- $\frac{1}{2}$ fermion with momentum $\mathbf{p}$ and spin $\mathbf{S}$, corresponding to left-handed (LH) and right-handed (RH) configurations, where the spin and momentum are anti-aligned and aligned, respectively [11].

Chirality plays a fundamental role in the weak interaction, which couples only to left-chiral fermions and right-chiral antifermions [7]. This behaviour implies that the weak interaction violates parity, in the sense that it is not invariant under a parity transformation (a spatial inversion, $\mathbf{x} \rightarrow -\mathbf{x}$), since such a transformation exchanges left- and right-handed states, as discussed in **Section 3.5.2**.

This is most clearly seen in neutrinos, for which only left-handed neutrinos and right-handed antineutrinos participate in weak processes. A direct experimental demonstration of this non-invariance under parity was provided by the Wu experiment, which observed an asymmetric angular distribution of electrons in β -decay relative to the nuclear spin [12]. For ultra-relativistic particles, chirality and helicity coincide, allowing the chiral structure of the weak interaction to be interpreted in terms of helicity.

The chiral structure of the weak interaction gives rise to a strong correlation between the muon spin direction and the angular distribution of its decay products. For the decay shown in **Eqn. 2**, high-energy positrons are preferentially emitted along the direction of the muon spin at the moment of decay. This asymmetry is energy-dependent, with high-energy positrons showing a strong correlation with the



muon spin, whilst the distribution becomes isotropic when averaged over all positron energies. This behaviour arises from the three-body nature of the decay, which produces a continuous spectrum of positron energies and leads to an energy-dependent correlation between the positron emission angle and the muon spin.

Thus, by measuring the time-dependent energy and angular distribution of the emitted positrons, the muon spin precession can be inferred. This spin–momentum correlation underpins precision measurements of the anomalous magnetic moment [6] and searches for a permanent EDM.

3.4 Discrete Symmetries

In **App. C**, Lorentz transformations describing continuous boosts and rotations are introduced. The Lorentz group also includes discrete spacetime transformations, such as parity $\mathbb{P}$ and time reversal $\mathbb{T}$. Together with the internal symmetry of charge conjugation $\mathbb{C}$, these operations underpin studies of fundamental symmetry violation in particle physics [8].

3.4.1 Parity Transformation

The parity transformation inverts all spatial axes whilst leaving the time coordinate unchanged. When the parity operator $\mathbb{P}$ acts on a spatial displacement vector $\mathbf{r}$, it reverses its direction, such that $\mathbb{P}\mathbf{r} = -\mathbf{r}$.

Vectors that change sign under parity are termed vectors (polar vectors), whilst those that remain unchanged are pseudovectors (axial vectors). The same transformation applies to other vectors, such as velocity $\mathbf{v}$ and linear momentum $\mathbf{p}$, with $\mathbb{P}\mathbf{v} = -\mathbf{v}$ and $\mathbb{P}\mathbf{p} = -\mathbf{p}$. For example, the angular momentum $\mathbf{L} = \mathbf{r} \times \mathbf{p}$ is invariant under parity. Since both $\mathbf{r}$ and $\mathbf{p}$ change sign, their cross product remains unchanged, $(-\mathbf{r}) \times (-\mathbf{p}) = \mathbf{r} \times \mathbf{p}$, and hence $\mathbf{L}$ transforms as a pseudovector.

Applying parity transformations to Maxwell’s equations shows that the electric field $\mathbf{E}$ behaves as a vector, whereas the magnetic field $\mathbf{B}$ transforms as a pseudovector. This distinction is fundamental to understanding how physical quantities respond to spatial inversion and underpins the analysis of symmetry violation in weak interactions.

3.4.2 Charge-Conjugation

Another discrete symmetry is the charge-conjugation operator, $\mathbb{C}$, which exchanges particles with their antiparticles. Its primary effect is to reverse the sign of all internal additive quantum numbers, such as electric charge, baryon number, and lepton number. Since electric and magnetic fields are sourced by charges and currents, they also change sign under charge conjugation, with $\mathbb{C}\mathbf{E} = -\mathbf{E}$ and $\mathbb{C}\mathbf{B} = -\mathbf{B}$.

In quantum field theory, $\mathbb{C}$ acts on field operators, transforming particles into antiparticles by reversing all internal quantum numbers. Charge conjugation therefore exchanges matter and antimatter. When considered alongside parity, charge conjugation symmetry was once believed to be universal. However, experiments on weak decays demonstrated that both $\mathbb{C}$ and $\mathbb{P}$ symmetries are violated in weak interactions [7]. This led to the study of the combined $\mathbb{CP}$ symmetry, which is also known to be violated.

3.4.3 Time-Reversal

The time-reversal operator, $\mathbb{T}$, is an anti-unitary operator that reverses the direction of time and performs complex conjugation of quantum states [8]. Its action can be written as $\mathbb{T}t = -t$ and $\mathbb{T}i = -i$, where the second relation reflects the anti-unitary nature of $\mathbb{T}$, corresponding to complex conjugation.

Under $\mathbb{T}$, spatial coordinates remain unchanged, whereas quantities involving time derivatives reverse sign, such that $\mathbb{T}\mathbf{r} = \mathbf{r}$, $\mathbb{T}\mathbf{v} = -\mathbf{v}$, and $\mathbb{T}\mathbf{p} = -\mathbf{p}$. Angular momentum therefore reverses direction under time reversal, since $\mathbf{p}$ changes sign, whilst $\mathbf{r}$ does not. Similarly, spin angular momentum $\mathbf{S}$ and magnetic



dipole moments μ change sign, with $\mathbb{T}S = -S$ and $\mathbb{T}\mu = -\mu$.

Most fundamental interactions in nature, such as electromagnetism and the strong force, are invariant under time reversal. However, the weak interaction is known to violate this symmetry. Importantly, a permanent EDM would also violate time-reversal symmetry, as it requires the dipole moment vector to remain aligned with the spin under $\mathbb{T}$, rather than reversing along with it. Thus, evidence of a non-zero EDM directly signals $\mathbb{T}$ and, by the CPT theorem, $\mathbb{CP}$ violation, as explained in **Section 3.5.1**.

3.5 CP Violation

3.5.1 CPT Theorem

With all of these discrete transformations in hand, one might initially expect each to be an exact symmetry of nature. However, this is not the case. Only the combined operation of $\mathbb{C}$, $\mathbb{P}$, and $\mathbb{T}$ is guaranteed to be a symmetry in any local, Lorentz-invariant quantum field theory. This result is known as the CPT theorem [7], and can be expressed in terms of the Hamiltonian as $\mathbb{CPT}H = H^\dagger$, where $H^\dagger = (H^*)^T$ denotes the Hermitian conjugate.

A direct consequence of this is that any violation of $\mathbb{CP}$ symmetry must be accompanied by a corresponding violation of $\mathbb{T}$ symmetry, such that overall CPT invariance is preserved. Conversely, any observed violation of CPT symmetry would imply a breakdown of Lorentz invariance or locality in the underlying theory.

3.5.2 CP Violation in the Standard Model

The combined $\mathbb{CP}$ transformation exchanges particles with their antiparticles whilst inverting spatial coordinates, with parity reversing helicity such that left-handed states are mapped to right-handed states and vice versa [8].

It is well established that the electromagnetic and strong interactions conserve $\mathbb{CP}$ symmetry to a high degree of precision. In contrast, the weak interaction violates this symmetry, treating left- and right-handed states differently.

The first experimental observation of $\mathbb{CP}$ violation was made in 1964 by Christenson, Cronin, Fitch, and Turlay in neutral kaon decays [10], with further effects later observed in B -meson systems [13]. Within the SM, these effects are encoded in the Cabibbo–Kobayashi–Maskawa (CKM) matrix [3], as discussed in **Section 3.1**.

Electric dipole moments provide a particularly sensitive probe of $\mathbb{CP}$ violation, as a non-zero EDM directly violates $\mathbb{T}$ symmetry and, via the CPT theorem, therefore also implies $\mathbb{CP}$ violation, making it a null test of these discrete symmetries.

3.5.3 The Sakharov Conditions

The generation of a baryon asymmetry in the early Universe requires three necessary conditions, first outlined by Andrei Sakharov [2] in 1967.

First, processes must violate baryon number conservation. Second, both $\mathbb{C}$ and combined $\mathbb{CP}$ symmetries must be violated, and by the CPT theorem, this implies a corresponding violation of $\mathbb{T}$. Third, these interactions must occur out of thermal equilibrium. In equilibrium, any process that produces a baryon asymmetry is balanced by its inverse, which acts to erase it. A departure from equilibrium is therefore required to prevent this cancellation and allow a net asymmetry to develop.

Within the SM, all three conditions can in principle be satisfied; however, the level of $\mathbb{CP}$ violation is highly suppressed.

3.6 Quantum Dipole Moments

In classical systems, EDMs arise from a spatial separation between positive and negative charges (see **App. E**). For fundamental particles such as the muon, however, a non-zero EDM would imply an intrinsic separation of charge within a point-like particle – a concept that cannot be described within a classical framework, as an object with no spatial extent cannot possess an internal charge distribution. Consequently, within classical physics there is no reasonable explanation as to how a fundamental particle could exhibit a permanent EDM.

To understand why this picture must be extended, we turn to quantum field theory [8]. In quantum systems, particles like the muon are never truly isolated, but instead constantly interact with the vacuum through the emission and reabsorption of virtual photons and other particles. Each of these contributions arises from an interaction term in the Lagrangian density, $\mathcal{L}_{\text{int}}$, and can be represented diagrammatically through Feynman diagrams, leading to small corrections to the particle's electromagnetic coupling.

The simplest interaction, known as the Born-level (tree-level) process describes the basic electromagnetic coupling between the muon and the photon. Quantum corrections arise when additional vertices and loops are introduced, corresponding to higher-order interactions that modify this coupling, as illustrated in **Fig. 4**.

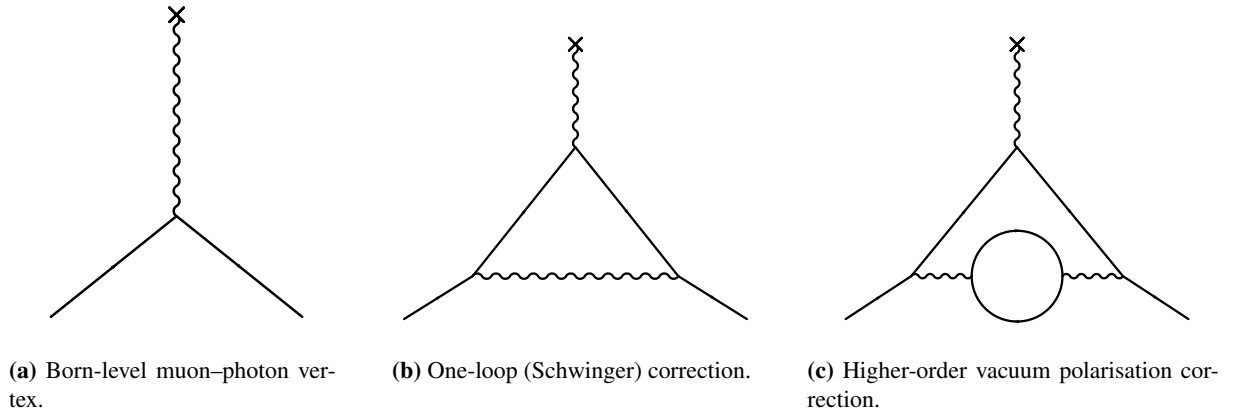


Figure 4: Successive quantum corrections to the muon–photon vertex: (a) Born-level, (b) one-loop (Schwinger) correction, and (c) higher-order vacuum polarisation. Solid lines represent fermions (e.g. the muon), whilst wavy lines represent photons. In each diagram, the continuous fermion line indicates the propagation of the fermion, whilst internal lines correspond to virtual particles with lifetimes constrained by the uncertainty principle.

These loop corrections modify the effective coupling between the muon and the electromagnetic field. Loops that conserve $\mathbb{CP}$ symmetry contribute only to the magnetic dipole term, leading to a small deviation from the Dirac prediction of $g = 2$. This deviation, known as the anomalous magnetic moment, was calculated to first order by Schwinger to be $a_\mu = (g - 2)/2 = \alpha/2\pi$ [14].

If, however, a virtual particle within a loop arises from a $\mathbb{CP}$ -violating interaction, such as those involving quarks via the weak interaction, as described by the CKM matrix [3], this symmetry is broken, giving rise to a small asymmetry between matter and antimatter contributions. This produces a small imbalance in the charge distribution along the muon spin axis, giving rise to an intrinsic EDM.

The presence of both magnetic and electric dipole moments modifies the precession of a particle's spin $\mathbf{S}$ in electromagnetic fields. This behaviour is described by the Bargmann–Michel–Telegdi (BMT) equation [15],

$$\frac{d\mathbf{S}}{dt} = (\boldsymbol{\omega}_a + \boldsymbol{\omega}_\eta) \times \mathbf{S}, \quad (4)$$

where $\boldsymbol{\omega}_a$ is the anomalous magnetic precession frequency and $\boldsymbol{\omega}_\eta$ represents an additional precession



arising from a non-zero EDM. The EDM-induced term can be written as:

$$\boldsymbol{\omega}_\eta = \frac{\eta q}{2m} \left(\frac{\mathbf{E}}{c} + \boldsymbol{\beta} \times \mathbf{B} \right), \quad (5)$$

with η parameterising the EDM strength. When considering the muon, we can relate this to the measurable quantity $|\mathbf{d}_\mu|$ via:

$$|\mathbf{d}_\mu| = \frac{\eta q \hbar}{4mc}. \quad (6)$$

The ω_a term governs the well-known $(g - 2)$ precession, whilst ω_η introduces an orthogonal precession component that tilts the muon spin out of the plane of motion. In the Fermilab $g-2$ experiment, this out-of-plane motion manifests as a vertical oscillation in the decay positron distribution, providing a direct experimental signature of a muon EDM. This vertical tilt is parametrised by an angle δ (**Fig. 5b**), which is encoded in the muon polarisation vector $\hat{\mathbf{s}}$ as:

$$\hat{\mathbf{s}} = \sin \phi \cos \delta \hat{\mathbf{i}} + \sin \phi \sin \delta \hat{\mathbf{j}} + \cos \phi \hat{\mathbf{k}}, \quad (7)$$

where $\phi = \omega t$ denotes the azimuthal phase around the storage ring. In the absence of an EDM ($\delta = 0$), the polarisation lies entirely within the plane of motion and no vertical component is expected. A non-zero EDM ($\delta \neq 0$) introduces a vertical component to the polarisation, leading to a measurable modulation in the decay angle between the muon and positron.

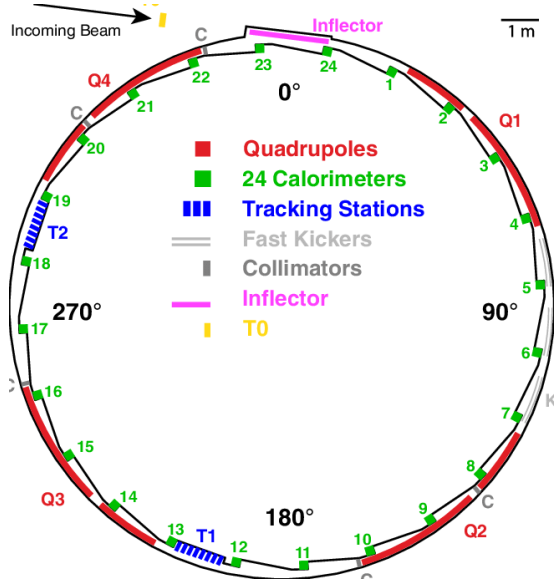
Within the SM, $\mathbb{CP}$ violation enters through the complex phase of the CKM matrix [3] and, in principle, the Pontecorvo–Maki–Nakagawa–Sakata (PMNS) matrix in the neutrino sector [16]. These effects induce a muon EDM through higher-order weak loops involving W bosons and neutrinos, but the resulting Standard Model prediction is exceedingly small, of order $|\mathbf{d}_\mu^{\text{SM}}| \sim 10^{-38} \text{ e} \cdot \text{cm}$, far below current experimental sensitivity. Theories beyond the SM (BSM), such as supersymmetry [17] or multi-Higgs doublet models [5], introduce new $\mathbb{CP}$ -violating phases that can enhance the EDM by many orders of magnitude. Consequently, a measurable muon EDM would provide clear evidence for new physics beyond the SM.

3.7 The Fermilab $g-2$ Experiment

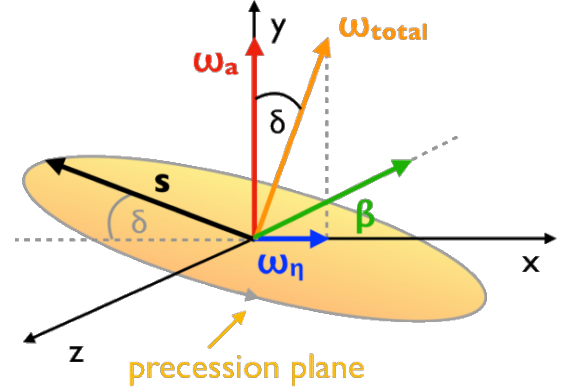
The Fermilab $g-2$ experiment measures the anomalous magnetic moment of the muon with unprecedented precision, whilst also providing sensitivity to a possible EDM. This arises from the transformation of the lab-frame magnetic field $\mathbf{B}$ into a combination of $\mathbf{B}'$ and $\mathbf{E}'$ fields in the muon rest frame (see **App. D**), with both effects encoded in the spin precession described by **Eqn. 4**.

The experiment stores polarised muons in a 14 m diameter storage ring (**Fig. 5a**) containing a highly uniform magnetic field of approximately 1.45 T [6]. As the muons circulate, their spins precess in the plane of motion at the anomalous frequency ω_a due to the $(g - 2)$ term. The stored muons decay according to **Eqn. 2**, and the emitted positrons are detected by an array of electromagnetic calorimeters distributed uniformly around the ring, as well as two tracking stations located at different azimuthal positions around the ring. High-energy positrons are preferentially emitted along the direction of the muon spin, causing the detected positron rate to oscillate in time at the spin precession frequency. By fitting this modulation, ω_a can be determined with high precision.

If the muon possesses a non-zero EDM, an additional precession term ω_η introduces a component of the spin motion out of the horizontal plane, as illustrated in **Fig. 5b**. The total precession frequency is then given by $|\boldsymbol{\omega}_{\text{total}}| = \sqrt{|\boldsymbol{\omega}_a|^2 + |\boldsymbol{\omega}_\eta|^2}$, resulting in a tilt of the spin precession away from the plane



(a) Schematic layout of the Fermilab Muon $g-2$ storage ring [18].



(b) Muon spin precession in the presence of a non-zero EDM in the muon rest frame, showing the tilt δ of the precession plane [19].

Figure 5: Overview of the Fermilab $g-2$ experiment and the effect of a non-zero EDM on the muon spin dynamics. A non-zero EDM induces a tilt δ in the spin precession away from the plane of motion.

of motion.

Although the presence of an EDM increases the magnitude of the precession frequency, $|\omega_{\text{total}}| > |\omega_a|$, the effect is negligible in practice since $|\omega_\eta| \ll |\omega_a|$, with $|\omega_\eta|/|\omega_a| \sim 10^{-3}$ [15, 20]. Consequently, the change in frequency is not experimentally accessible, and the tilt of the precession plane provides the most sensitive observable. This manifests as a time-dependent vertical asymmetry in the positron decay distribution.

The absence of such a signal places an upper bound on the muon EDM. This approach was used by the Brookhaven National Laboratory (BNL) to obtain a limit of $|d_\mu| < 1.5 \times 10^{-19} \text{ e} \cdot \text{cm}$ at the 90% confidence level [20].

4 Methods

4.1 Analysis Overview

To perform this analysis, the fractional energy of the decay positrons in the lab-frame, $\lambda = E/E_{\text{max}}$, the longitudinal² positron decay angle θ_L , and the associated probability density Z were evaluated at 29 discrete time steps. These time steps are labelled using the fractional precession period $T_i \equiv t_i/T = i/29$, with $i = 0, \dots, 28$, where T is the time taken for the spin to complete a full 2π precession. At each time step, the (θ_L, λ) phase space is discretised into a 500×500 grid, with the probability density $Z(\theta_L, \lambda; T_i)$ evaluated in each bin, for each time step.

The data used in this analysis are generated from analytic PDFs derived from the Michel decay parameters at Born-level, neglecting higher-order corrections [21]. Positrons are treated in the ultra-relativistic limit ($E \gg m_e c^2$), such that $v \approx c$. To ensure the validity of this approximation, the analysis is restricted to the range $\lambda \in [0.2, 1.0]$, as at lower energies the positron mass can no longer be neglected.

²Here, “longitudinal” refers to the direction defined by the lab-frame magnetic field, rather than the angle between the muon and positron momenta.

With these definitions, the (θ_L, λ) phase space can be visualised as a two-dimensional heatmap, with the probability density represented by a colour scale. Within this model, the largest EDM–no-EDM differences occur around T_7 and T_{22} , where the muon spin is maximally transverse to the momentum direction, corresponding to phases of $\pm\pi/2$. The phase-space distribution at T_7 is shown in **Fig. 6** for both the EDM and no-EDM cases.

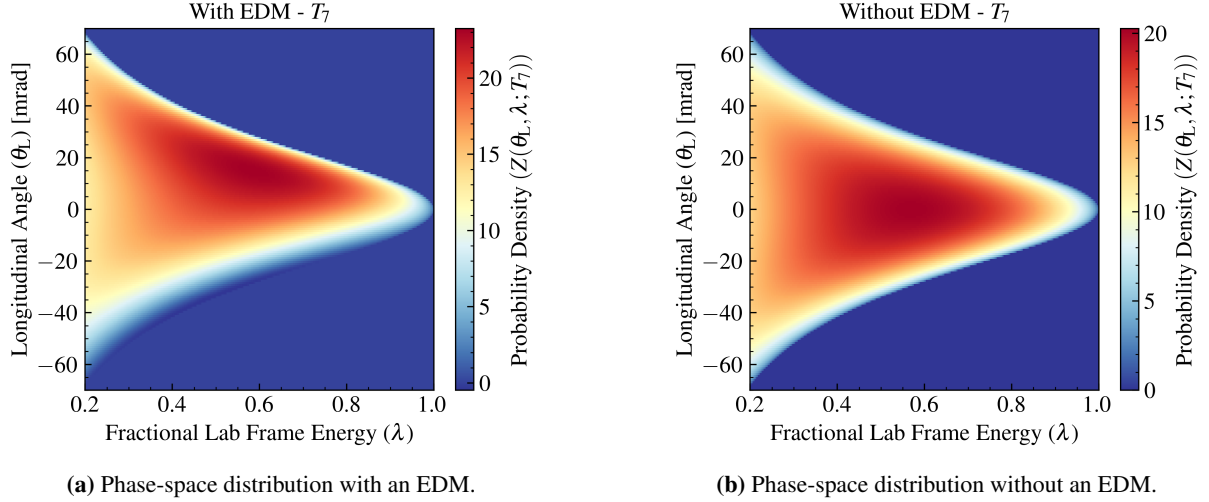


Figure 6: Two-dimensional (θ_L, λ) phase space at T_7 , with the probability density $Z(\theta_L, \lambda; T_7)$ represented by the colour scale. The EDM introduces a small θ_L asymmetry in the probability densities. Note that the data used to generate these plots used an extreme value of $\delta = 1.2$ rad for visualisation purposes. The distribution is most densely populated at small θ_L , reflecting the underlying decay kinematics, whilst lower-energy positrons exhibit a broader angular spread.

The EDM sensitivity depends on both the fractional energy of the emitted positrons and the longitudinal decay angle, as illustrated in **Fig. 7**, where the variation in oscillation amplitude with λ shows that the observable response to an EDM differs across energy, and hence that different regions of phase space contribute unequally to the sensitivity. To quantify this dependence and identify the regions of phase space with the greatest sensitivity, a figure of merit (FOM) is introduced, where larger values correspond to increased sensitivity to the muon EDM. To enable optimisation across the (θ_L, λ) grid, an arbitrary weighting function $w(\theta_L, \lambda)$ is defined, allowing specific regions of phase space to be emphasised or suppressed, and thereby modifying the effective statistical contribution of different regions to the measurement. For notational simplicity, the dependence on $w(\theta_L, \lambda)$ is left implicit unless required for clarity.

For a given energy fraction, we define the FOM as:

$$\text{FOM}(\lambda) = \sum_{i=0}^{28} \frac{(\langle \theta_L \rangle_{i,\text{EDM}} - \langle \theta_L \rangle_{i,\text{no-EDM}})^2}{\sigma_{i,\text{EDM}}^2 + \sigma_{i,\text{no-EDM}}^2}. \quad (8)$$

Here $\langle \theta_L \rangle_i$ is the mean longitudinal angle in the time bin i , and σ_i denotes the standard error on the mean of $\langle \theta_L \rangle_i$ in bin i (i.e. $\sigma_i = s_i / \sqrt{N_i}$, where s_i is the sample standard deviation and N_i represents the number of muons in the bin). The subscripts “EDM” and “no-EDM” denote quantities evaluated using datasets generated with and without an EDM, respectively. This form of the FOM quantifies the squared separation between the EDM and no-EDM datasets, normalised by the combined statistical uncertainty, and therefore represents the statistical distinguishability of the two cases. Varying N_i does not change the central value of the observable or the resulting FOM, but only rescales the statistical uncertainty.

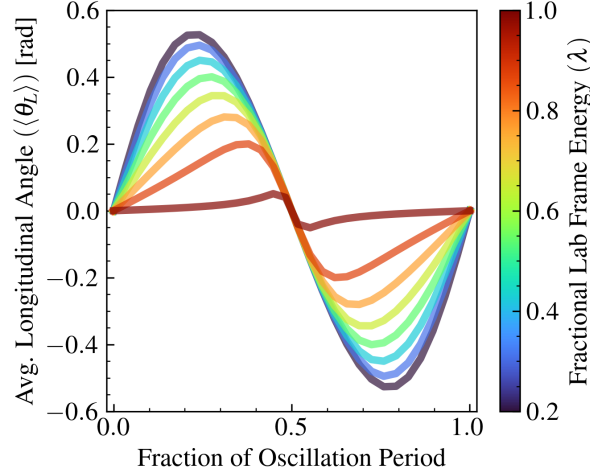


Figure 7: The average longitudinal angle plotted against the fraction of the total oscillation period. As the colour varies from dark blue to red, the fractional lab-frame energy λ increases from 0.2 to 1.0. Higher-energy positrons exhibit a narrower angular distribution, whilst lower-energy positrons display a broader distribution.

Several simplifying assumptions can be made to simplify this expression. In the absence of an EDM, the model is constructed such that the expected mean longitudinal angle is zero, since no-EDM signal implies no vertical component to the precession. Furthermore, the two samples are generated with identical statistics, and the presence of an EDM is assumed to shift $\langle \theta_L \rangle$ without modifying the underlying variance of the distribution, such that $\sigma_{i,\text{no-EDM}}^2 = \sigma_{i,\text{EDM}}^2$ for each value of i . With these assumptions, we can simplify our equation for the FOM to obtain:

$$\text{FOM}(\lambda) = \sum_{i=0}^{28} \frac{(\langle \theta_L \rangle_{i,\text{EDM}})^2}{2\sigma_{i,\text{EDM}}^2}, \quad (9)$$

which provides a one-dimensional measure of EDM sensitivity at fixed λ . This one-dimensional FOM can be used to identify the optimal fractional energy for EDM sensitivity. However, it is also useful to define a FOM across the full two-dimensional phase space. This is obtained by summing over the N_λ discrete energy bins labelled by λ_k ,

$$\text{FOM} = \sum_{k=1}^{N_\lambda} \text{FOM}(\lambda_k). \quad (10)$$

Having defined the FOM, this provides a single metric that captures the combined sensitivity across the full (θ_L, λ) grid, and is used to assess the performance of a given weighting scheme. The optimisation is performed using the one-dimensional FOM at fixed λ , allowing the weighting to be determined independently in each energy slice, with the full two-dimensional FOM used to evaluate the overall performance.

4.2 Estimations of $\langle \theta_L \rangle_i$ and σ_i^2

Each bin is labelled by a time index i and an angle index j , with population p_{ij} and corresponding weight w_{ij} . In the weighted case, each bin contributes to the sums over θ_L with an effective weight $w_{ij}p_{ij}$, thereby modifying the relative statistical contribution of different regions of phase space to the observable.

If w_{ij} is constant across all bins, i.e. uniform over phase space, this reduces to the unweighted case, consistent with the scale invariance of the FOM under an overall rescaling of the weights.



We can now define the weighted mean longitudinal angle in time bin i as:

$$\langle \theta_L \rangle_i = \frac{\sum_j p_{ij} w_{ij} \theta_{L,ij}}{\sum_j p_{ij} w_{ij}}. \quad (11)$$

This corresponds to the expectation value of the longitudinal angle under the weighted distribution, with the weighting function effectively redefining the sampling of phase space. The corresponding weighted variance is given by:

$$s_i^2 = \frac{\sum_j p_{ij} w_{ij} (\theta_{L,ij} - \langle \theta_L \rangle_i)^2}{\sum_j p_{ij} w_{ij}}. \quad (12)$$

Finally, the standard error on the mean is defined as:

$$\sigma_i = \frac{s_i}{\sqrt{n_{\text{eff},i}}}, \quad (13)$$

where $p_{ij} = NZ_{ij}$ represents the effective population in the j th (θ_L, λ) bin at time step i , with Z_{ij} expressed in radians to ensure consistency in the integration over θ_L . The parameter N sets the overall event scale and can be chosen arbitrarily at this stage, as it simply rescales the bin populations without affecting the relative weighting or the resulting figure of merit. For consistency, it is fixed to 10^5 throughout the analysis.

Once weighting is introduced, the statistical contribution of each bin is no longer uniform, and the simple scaling with N alone no longer characterises the uncertainty. To account for this, we introduce an effective sample size $n_{\text{eff},i}$, defined as:

$$n_{\text{eff},i} = \frac{\left(\sum_j p_{ij} w_{ij}\right)^2}{\sum_j p_{ij} w_{ij}^2}. \quad (14)$$

This quantity represents the effective number of independent events contributing to the weighted estimate, accounting for the increased variance introduced by non-uniform weighting. Using the definitions above, $\langle \theta_L \rangle_i$ and σ_i^2 are evaluated for both the EDM and no-EDM datasets at each time step i and for each energy bin λ_k , providing the inputs required to compute the figure of merit for a given fractional energy and weighting scheme. These quantities are then combined to compute the two-dimensional figure of merit for a given choice of weighting function $w(\theta_L, \lambda)$, forming the basis of the optimisation procedure described in the following sections.

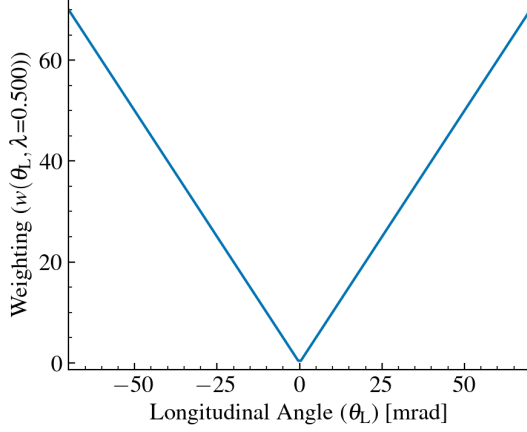
4.3 Empirical Weighting Schemes

To guide the construction of an optimal weighting function, an initial heuristic approach was considered. A natural first approach is to upweight larger values of the longitudinal angle, as the EDM-sensitive contribution is expected to increase with $|\theta_L|$. This weighting is illustrated in **Fig. 8**.

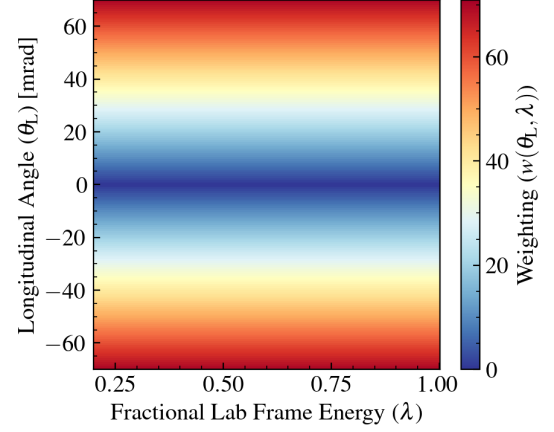
A more effective empirical approach is to weight regions according to the difference between the EDM and no-EDM distributions, such that regions with the greatest asymmetry are preferentially enhanced. This is illustrated in **Fig. 9**. This approach directly targets regions of phase space where the EDM induces the largest observable deviations, and therefore provides a closer approximation to the underlying sensitivity structure.

This weighting is independent of the particular time step chosen (provided $T_i \neq T_0$). Across the

ring, the regions where Z_{EDM} and $Z_{\text{NO EDM}}$ differ do not change with T_i ; only the overall magnitude of the difference varies. Since the figure of merit is invariant under an overall rescaling of the weights, this change in magnitude does not affect the resulting weight map. The only exception is $T_i = T_0$, for which the difference vanishes by construction.

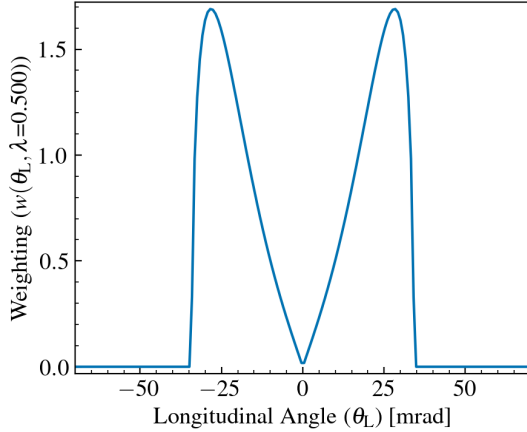


(a) One-dimensional slice of the weighting at $\lambda = 0.500$, showing w increasing with $|\theta_L|$.

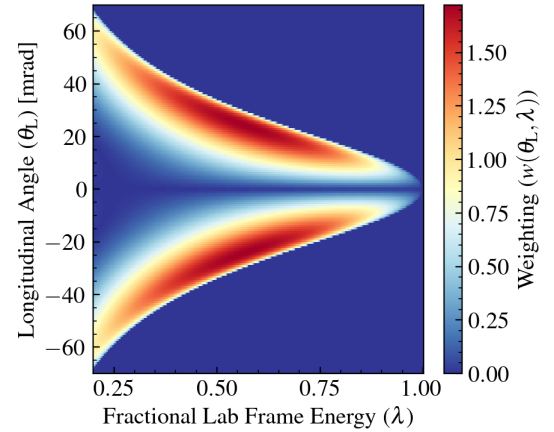


(b) Full two-dimensional weight map over the (θ_L, λ) grid.

Figure 8: Illustration of the heuristic weighting $w(\theta_L, \lambda) = |\theta_L|$: (a) shows a slice at $\lambda = 0.500$, and (b) shows the weight map over the full range.



(a) One-dimensional slice of the weighting at $\lambda = 0.500$, showing w varying to emphasise the sections with the largest difference between the EDM and no-EDM signals.



(b) Full two-dimensional weight map over the (θ_L, λ) grid.

Figure 9: Illustration of the heuristic weighting $w(\theta_L, \lambda) = |Z_{\text{EDM}} - Z_{\text{NO EDM}}|$: (a) shows a slice at $\lambda = 0.500$, and (b) shows the weight map over the full range, zoomed in on the range of angles which have a non-zero weight.

Whilst the difference-based weighting captures the dominant structure of the EDM sensitivity, it does not account for the variation in event density across phase space, and is therefore not guaranteed to be optimal. This motivates the need for a systematic optimisation procedure to determine the weighting that maximises sensitivity, which is introduced in the following section.

4.4 Weighting Optimisation with CMA-ES

The initial heuristic weighting schemes demonstrated a clear improvement in sensitivity, motivating a systematic optimisation of the weighting function. This was performed using the Covariance Matrix



Adaptation Evolution Strategy (CMA-ES), a derivative-free optimiser well suited to high-dimensional, black-box problems, where the objective function is non-analytic and gradient information is not readily available [22].

The objective function is given by the one-dimensional FOM at fixed λ , with the weighting function parameterised over the θ_L grid. Due to the high dimensionality of the full (θ_L, λ) space, the optimisation was performed independently for fixed values of λ , with the resulting weights combined via linear interpolation to construct the full two-dimensional map. This decomposition allows the optimisation to focus on the dominant variation in θ_L at fixed energy, whilst maintaining computational tractability across the full phase space. To reduce the dimensionality of the optimisation (**App. F**) and restrict it to physically relevant regions, two constraints were imposed.

First, the longitudinal angle was restricted to the kinematically allowed region $|\theta_L| \leq \theta_{\max}(\lambda)$, where the maximum angle is defined as $\theta_{\max}(\lambda) = \sin^{-1} \left(\frac{\sqrt{\lambda(1-\lambda)}}{\gamma\lambda} \right)$ [21]. Outside this region, the probability density vanishes identically, and therefore such regions do not contribute to the figure of merit, and the weights were set to zero.

Secondly, the weighting function was required to be symmetric in θ_L , such that $w(-\theta_L, \lambda) = w(\theta_L, \lambda)$. This reflects the underlying symmetry of the unperturbed decay distribution in θ_L , ensuring that the weighting does not introduce artificial asymmetries unrelated to the EDM signal, and also reduces the number of parameters, ensuring that the optimiser focuses on physically meaningful structure.

With these constraints, each optimisation involves $\mathcal{O}(10^2)$ parameters rather than the full $\mathcal{O}(10^5)$, allowing the CMA-ES algorithm to efficiently explore the parameter space without becoming trapped in an intractably large search space. The optimisation problem remains high-dimensional and non-analytic, with no accessible gradient information, rendering gradient-based methods unsuitable. Alternative approaches, such as Bayesian optimisation, do not scale efficiently to the dimensionality considered here. CMA-ES is therefore well suited, as it adaptively samples candidate solutions from a multivariate distribution, enabling efficient exploration of high-dimensional, non-convex parameter spaces.

Having established this optimisation framework, the resulting weights can now be analysed and compared to an analytic construction motivated by the observed structure in the figure of merit.

4.5 Analytic Stationarity Weight

To derive the weighting that extremises the one-dimensional FOM, we begin by recasting it as a functional of the weight $w(\theta_L; \lambda)$:

$$\text{FOM}[w(\theta_L; \lambda)] = \sum_{i=0}^{28} \frac{(\langle \theta_L \rangle_{i,\text{EDM}})^2}{2\sigma_{i,\text{EDM}}^2}, \quad (15)$$

where λ is treated as a fixed parameter, such that an independent optimal weight is obtained for each $\lambda \in [0.2, 1]$. This reduces the optimisation to a one-dimensional problem in θ_L at fixed energy. The optimal weighting is then obtained by imposing the stationarity condition,

$$\frac{\delta \text{FOM}}{\delta w(\theta_L; \lambda)} = 0, \quad (16)$$

where $\delta/\delta w$ denotes a functional derivative, as defined in **App. G**. For tractability, the derivation is performed in the continuous limit with respect to θ_L , whilst retaining the discrete sum over time bins i . The resulting weight is therefore a self-consistent, pointwise analytic solution that extremises the



figure of merit, providing a direct link between the structure of the optimal weighting and the underlying statistical sensitivity.

4.6 EDM Extraction Procedure

For each fixed λ , the time dependence of the weighted mean angle $\langle\theta_L\rangle_i$, which encodes the EDM-induced oscillatory signal, is fitted using `scipy.optimize.curve_fit` [23]:

$$\langle\theta_L\rangle_i = \frac{A(\lambda) \sin\left(\frac{2\pi i}{29}\right)}{1 + B \cos\left(\frac{2\pi i}{29}\right)}, \quad (17)$$

where $i = 0, \dots, 28$ labels the discrete time bins over one full spin oscillation. This functional form captures the expected oscillatory behaviour of the observable in the laboratory frame [21].

The extracted values of $A(\lambda)$ are then fitted as a function of λ using the analytic form:

$$f(\lambda) = C \left(\frac{(\lambda - 1)(2\lambda + 1)}{4\lambda^2 - 5\lambda - 5} \right), \quad (18)$$

which follows from the laboratory-frame decay kinematics [21]. The fitted parameter $C \pm \Delta C$ corresponds to the EDM-induced tilt in the laboratory frame, providing a direct measure of the EDM-induced modification to the spin precession.

The corresponding rest-frame tilt is obtained via a Lorentz boost, $\delta = \gamma C$, with uncertainty $\Delta\delta = \gamma \Delta C$ determined through standard error propagation. As the datasets are generated from analytic probability densities, this procedure represents an idealised limit in which statistical fluctuations are absent, isolating the intrinsic sensitivity of the method.

This extraction applies directly to the unweighted observables. However, upon introducing the weighting scheme, the fitted amplitude $A(\lambda)$ no longer follows the analytic form given by **Eqn. 18**, as the weighting modifies the observable. Rather than re-deriving this relation, an empirical calibration is performed.

The EDM contribution is isolated by taking the difference between datasets generated with and without an EDM, and rescaled by a factor α before being added back to the without-EDM dataset, producing an effective EDM strength $\alpha\delta$:

$$\text{withEDM}_{\text{scaled}} = \alpha(\text{withEDM} - \text{withoutEDM}) + \text{withoutEDM}. \quad (19)$$

By varying α , the EDM-dependent component can be continuously scaled, allowing the relationship between the fitted amplitude and δ to be reconstructed for the weighted observables. A likelihood-based extraction of δ is also performed as an independent cross-check, as described in **App. H**.

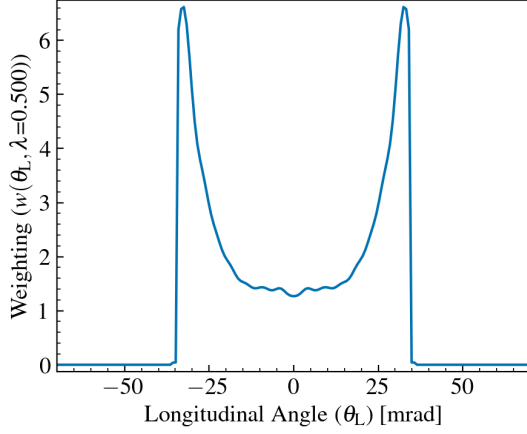
5 Results

5.1 Observable Behaviour

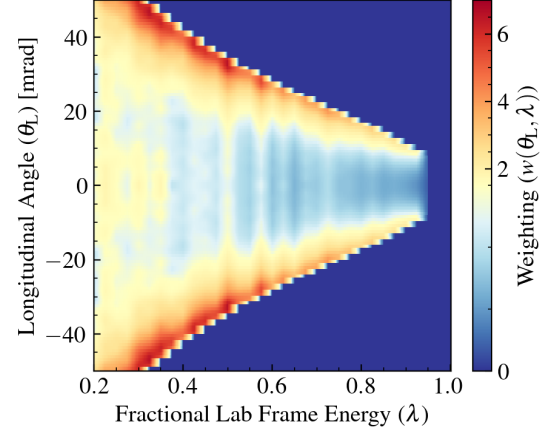
The average longitudinal angle $\langle\theta_L\rangle$ is calculated for each time step and plotted against the fractional precession period for a range of fractional lab-frame energies λ , as shown in **Fig. 7**. The observable exhibits the expected oscillatory behaviour arising from the precession of the muon spin. The amplitude of these oscillations shows a strong dependence on λ , with different energy regions producing markedly different responses to the EDM-induced signal. Since the EDM is correlated to this observable, this variation in response directly translates into a non-uniform sensitivity across λ . This demonstrates that

information about the muon EDM, manifesting as the tilt δ , is not distributed uniformly across phase space. Consequently, treating all events equally is suboptimal, further motivating the introduction of a weighting function.

5.2 Structure of the Optimised Weight



(a) One-dimensional slice of the optimised weighting at $\lambda = 0.500$, showing a symmetric structure in θ_L with maxima at the largest kinematically allowed values of $|\theta_L|$ and suppression near $\theta_L \approx 0$.



(b) Optimised weighting function over the (θ_L, λ) phase space, showing enhanced weighting at large $|\theta_L|$ and low λ , with boundaries defined by the kinematic limits.

Figure 10: Structure of the optimised weighting function $w(\theta_L, \lambda)$ obtained using CMA-ES. The weighting is concentrated at large angles and lower fractional energies, whilst regions near $\theta_L \approx 0$ and high λ are suppressed.

An optimised weight is obtained using CMA-ES, calculated for 32 individual λ slices. These slices are then combined via linear interpolation to construct a continuous weighting scheme, as shown in **Fig. 10b**, with a representative slice at $\lambda = 0.500$ shown in **Fig. 10a**.

The optimised weighting assigns the largest weights to regions of large $|\theta_L|$ within the kinematically allowed phase space, whilst suppressing contributions from central angles without eliminating them entirely. The weight is symmetric in θ_L by construction and exhibits small fluctuations around a minimum at low $|\theta_L|$.

At higher values of λ , this suppression becomes more pronounced in the central region, whilst the weighting at larger $|\theta_L|$ is constrained by the kinematic limits. This behaviour does not follow a simple monotonic dependence on λ , and instead varies across the kinematically allowed phase space.

The weighting map does not appear perfectly smooth across λ , exhibiting small-scale variations between neighbouring slices. This arises from the independent optimisation performed at each value of λ , with the full map constructed by combining these slices via interpolation, rather than reflecting any underlying physical structure. Importantly, these variations do not affect the dominant features of the weighting, which remain clearly defined. The dominant features of the weighting are therefore robust against these interpolation artefacts.

Overall, the optimiser enhances regions with the largest observable response, whilst suppressing those with weaker contributions. The resulting structure is stable across λ and consistent with the behaviour of the figure of merit, motivating a comparison with the analytic form.

5.3 Comparison of Analytic and Numerical Weighting

Following the procedure outlined in **App. G**, the analytic weight is found to satisfy:

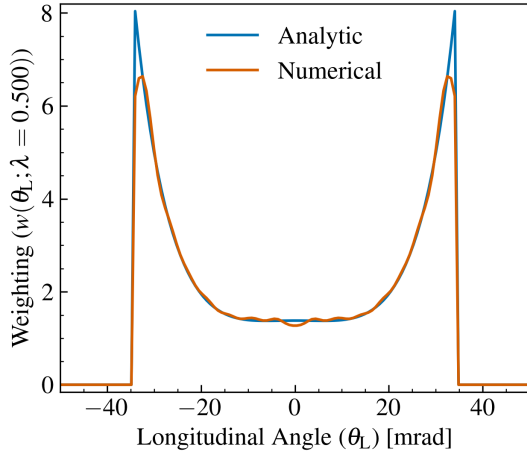
$$w(\theta_L; \lambda) = \frac{\sum_i \frac{\langle \theta_L \rangle_i p_i(\theta_L; \lambda) a_i(\theta_L)}{\sigma_i^2}}{\sum_i \frac{\langle \theta_L \rangle_i p_i(\theta_L; \lambda) b_i}{\sigma_i^2}}, \quad (20)$$

where we define:

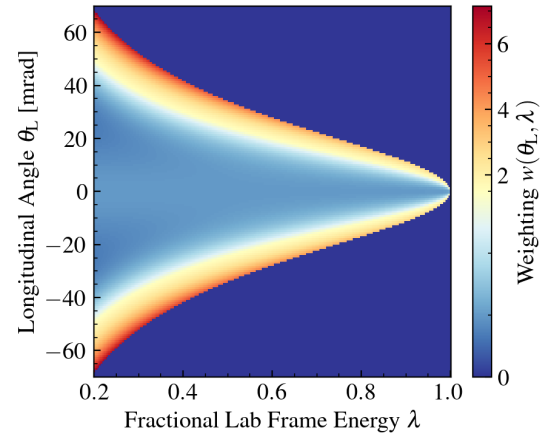
$$a_i(\theta_L) = (\theta_L - \langle \theta_L \rangle_i) - \frac{\langle \theta_L \rangle_i}{2\sigma_i^2}(\theta_L - \langle \theta_L \rangle_i)^2 + \frac{3\langle \theta_L \rangle_i}{2}, \quad b_i = \frac{\int p_i(\theta_L; \lambda) w(\theta_L; \lambda) \theta_L d\theta_L}{\int p_i(\theta_L; \lambda) w(\theta_L; \lambda)^2 d\theta_L}. \quad (21)$$

Although this expression is implicit in w , as it appears on both sides of the equation, it can be evaluated by supplying a trial weighting on the right-hand side. This yields a well-defined analytic weighting for a given initial choice, with the resulting solution found to be stable and only weakly dependent on this choice. In practice, repeated application of this mapping does not lead to stable convergence, and is therefore not used as an iterative scheme.

Evaluating this expression yields a weighting scheme that satisfies the stationarity condition (**Eqn. 16**). The full weighting scheme, along with a representative slice, is shown in **Fig. 11**. Both the numerical and analytic approaches exhibit similar structural features, enhancing regions near the kinematic boundary, whilst suppressing contributions from central angles without reducing them to zero.



(a) One-dimensional slice of the analytic weighting (blue) and the numerical (orange) weighting at $\lambda = 0.500$, both exhibiting a symmetric structure in θ_L with maxima at the largest kinematically allowed values of $|\theta_L|$ and suppression near $\theta_L \approx 0$. The agreement between the two is close.



(b) Analytic weighting function over the (θ_L, λ) phase space, showing enhanced weighting at large $|\theta_L|$ and low λ , with boundaries defined by the kinematic limits. The resulting structure is smooth across λ .

Figure 11: Structure of the analytic weighting function $w(\theta_L, \lambda)$ obtained from **Eqn. 20**. The weighting is concentrated at large angles and lower fractional energies, whilst regions near $\theta_L \approx 0$ and high λ are suppressed.

From the one-dimensional slice at $\lambda = 0.500$ in **Fig. 11a**, the close agreement between the numerical and analytic weights can be observed. To quantify this agreement, we compute the cosine similarity, defined between two weight vectors $\mathbf{w}_1$ and $\mathbf{w}_2$ as:

$$\text{cosine similarity} = \frac{\mathbf{w}_1 \cdot \mathbf{w}_2}{\|\mathbf{w}_1\| \|\mathbf{w}_2\|}. \quad (22)$$



The cosine similarity between the numerical and analytic weighting schemes at $\lambda = 0.500$ is found to be 0.996, indicating near-perfect structural alignment, with a value of 1 corresponding to exact agreement. This shows that the numerical optimiser converges to a weighting structure consistent with the analytic solution.

Having established this structural agreement, the performance of the weighting schemes can now be evaluated using the figure of merit.

5.4 Figure of Merit Performance

The one-dimensional figure of merit is evaluated as a function of λ for each weighting scheme, with the results shown in **Fig. 12**. The figure of merit was computed for the unweighted case, the difference based heuristic weighting, the CMA-ES optimised weighting, and the analytic weighting.

Relative to the unweighted baseline, the heuristic approach provides a 22.6% improvement in EDM sensitivity. The CMA-ES and analytic weighting schemes yield improvements of 49.3% and 49.1%, respectively. This corresponds to an increase in effective statistical power.

These results demonstrate that weighting enhances sensitivity to the muon EDM, with the optimised schemes outperforming the heuristic approach. The close agreement between the analytic and CMA-ES weightings is consistent with the analytic form capturing the same underlying structure.

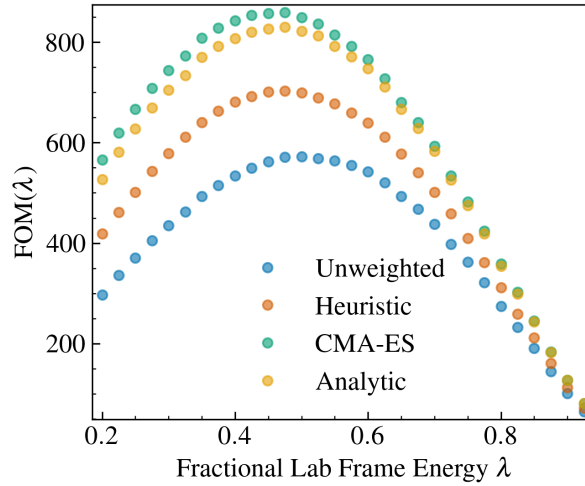
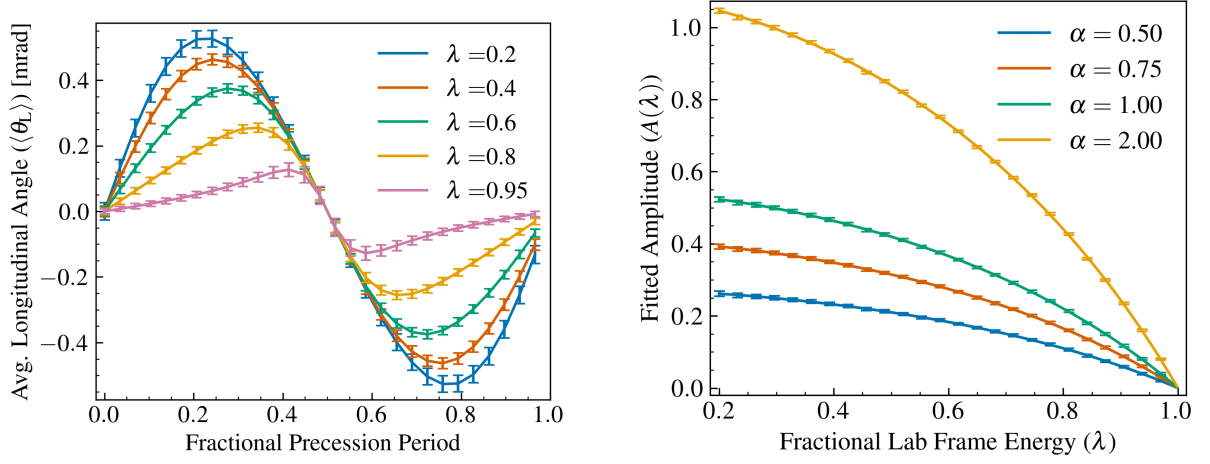


Figure 12: Comparison of the figure of merit for unweighted, heuristic, and optimised weighting schemes as a function of λ . The optimised weightings provide a substantial enhancement in sensitivity relative to the unweighted case.

5.5 Extraction of the EDM Parameter δ

The time evolution of the average longitudinal angle $\langle \theta_L \rangle$ over a single precession period is shown in **Fig. 13a** for representative values of λ . A clear oscillatory behaviour is observed for all λ , with a λ -dependent amplitude.

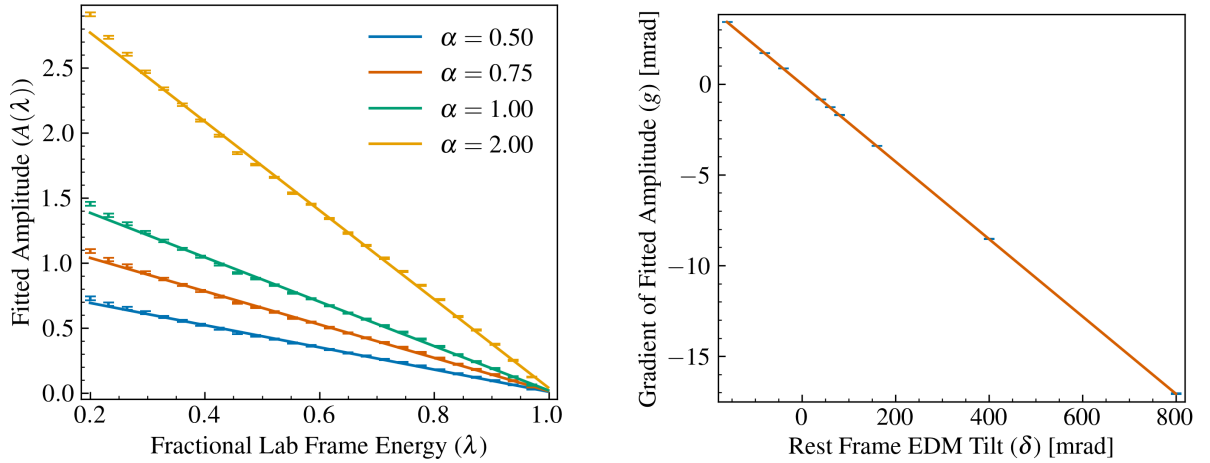
In the unweighted case, the extracted value is found to be $\delta = (80.0 \pm 0.1)$ mrad, consistent with the injected value $\delta_0 = 80$ mrad within numerical precision. In the weighted case, however, the functional form of the amplitude changes, and the previous extraction method is no longer directly applicable. Instead, the amplitude exhibits an approximately linear dependence on λ , as demonstrated for a range of EDM strengths scaled using the previously defined parameter α , as per **Eqn. 19**.



(a) Time evolution of $\langle \theta_L \rangle$ over a single precession period for representative values of λ .

(b) Fitted amplitude for the unweighted average angle as a function of λ , following **Eqn. 18**. The uncertainties on $A(\lambda)$ are of $\mathcal{O}(10^{-2})$.

Figure 13: Unweighted behaviour of the observable as a function of time. The oscillatory time evolution motivates the extraction of the amplitude, which follows the expected analytic dependence on λ . The fitted amplitudes from (a) are plotted against λ in (b) for a range of EDM strengths.



(a) Fitted amplitude for the weighted average angle as a function of λ , which becomes approximately linear. The uncertainties on $A(\lambda)$ are of $\mathcal{O}(10^{-2})$.

(b) Gradient of the weighted amplitude plotted against the rest-frame EDM tilt δ , showing a linear calibration, where $m = (-0.021334 \pm 0.000002)$ and $c = (-0.0002 \pm 0.0007)$ mrad are the fitted parameters. The uncertainties on g are of $\mathcal{O}(10^{-2})$.

Figure 14: Weighted behaviour of the observable. The application of the optimal weighting transforms the amplitude into a linear function of λ , establishing a linear relationship between the gradient and the EDM tilt.

The linear behaviour observed in the weighted case indicates that the EDM dependence is captured by the gradient of the amplitude. Defining the gradient as $g \equiv dA/d\lambda$, this relationship can be investigated by plotting g as a function of the injected rest-frame tilt δ . As shown in **Fig. 14b**, a clear linear relationship is observed between the gradient of the weighted amplitude and the rest-frame EDM tilt, with a reduced $\chi^2 = 0.1$, indicating that the linear model provides a good description of the data. This behaviour was verified over an extended range of $\alpha \in [-2.0, 10.0]$, confirming that the linear relation persists across a wide range of injected EDM strengths. This linear calibration provides a means to



extract the EDM tilt via:

$$\delta = \frac{g - c}{m}, \quad (23)$$

where $m = (-0.021334 \pm 0.000002)$ and $c = (-0.0002 \pm 0.0007)$ mrad are the resulting parameters from the fit. As a demonstration, the gradient obtained for the $\alpha = 1$ case yields $\delta = (80.0 \pm 0.1)$ mrad, which is consistent with the injected value. The uncertainty on δ is obtained by propagating the errors on the fitted gradient and calibration parameters.

The application of the optimal weighting results in an observable for which the EDM dependence is approximately linear, enabling the extraction of δ using the calibrated relationship. This behaviour is consistent with the trends observed in the figure of merit.

5.6 Log-Likelihood Cross-Check

This subsection presents an independent validation of the extracted value of δ , following the procedure outlined in **App. H**. The multinomial log-likelihood was evaluated as a function of δ , with the best-fit value determined from the minimum of the distribution and the corresponding uncertainty.

Values of δ were obtained for datasets generated with and without an EDM signal. For the dataset without an EDM, the result was $\delta = (0.0 \pm 0.9)$ mrad, consistent with the expected null result. For the dataset with an EDM, the result was $\delta = (80.0 \pm 0.9)$ mrad, in agreement with the injected value $\delta_0 = 80$ mrad.

The extracted value in the EDM case is also consistent with the result obtained using the gradient-based method in **Section 5.5**, providing an independent cross-check of the extraction procedure. This agreement indicates consistency between the two extraction methods.

6 Discussion

6.1 Interpretation of the Optimal Weight

As shown in **Fig. 11b**, the analytic optimal weighting assigns the largest weights to regions near the kinematic boundary at large $|\theta_L|$, whilst suppressing, but not eliminating, the central region around $\theta_L \approx 0$. This behaviour can be understood in terms of how the EDM modifies the observable. A non-zero EDM introduces a vertical tilt δ to the muon spin precession (**Eqn. 7**), which manifests as an asymmetry in the angular distribution of the decay positrons. The sensitivity to the EDM is therefore determined by how strongly the observable responds to this tilt.

Regions of large $|\theta_L|$ provide greater sensitivity, as variations in the muon spin direction produce larger changes in the observed positron angle. In contrast, near $\theta_L \approx 0$, this response is suppressed, leading to reduced sensitivity. This behaviour is consistent with **Fig. 6b**, where the central region contains the highest density of events but exhibits a smaller spread in angle, whilst larger angles are less populated but show a greater variation. This follows directly from the structure of the figure of merit (**Eqn. 9**), which favours regions where the signal, characterised by $\langle \theta_L \rangle$, is large relative to the uncertainty on the mean, σ . This uncertainty depends both on the spread of the angular distribution and the effective number of events. Regions at large $|\theta_L|$ exhibit greater variation in θ_L , enhancing the signal, but are statistically suppressed due to lower event rates. In contrast, the central region contains a large number of events, reducing the uncertainty on the mean, but exhibits a smaller angular variation and therefore a weaker signal.

The optimal weighting therefore reflects a balance enforced by the figure of merit: large angles enhance the signal term through increased variation, whilst the central region reduces the uncertainty on



the mean through its higher event rate. Consequently, the weighting retains a non-zero contribution from small angles rather than suppressing them entirely.

The dependence on λ arises from the decay kinematics of the positrons. At higher fractional energies, positrons are preferentially emitted along the muon spin direction, strengthening the correlation between the observable and the spin orientation. This enhances the sensitivity to an EDM-induced tilt. At lower energies, the emission becomes more isotropic, reducing this effect and leading to a weaker dependence on δ .

The resulting weighting can therefore be understood as selecting regions of phase space that maximise sensitivity to the EDM, whilst retaining sufficient statistical precision, effectively optimising the information extracted from the observable.

6.2 Interpretation of the Analytic Weight

As seen in **Fig. 11a**, the CMA-ES and analytic weights are visually very similar, with this agreement quantified by their high cosine similarity. This indicates that the optimiser is identifying structure in the underlying sensitivity, rather than converging to an arbitrary numerical solution.

The analytic weighting is obtained by extremising the sensitivity through a stationarity condition, and is therefore determined by how the observable depends on δ . The numerical optimisation is effectively solving the same problem, searching for the weighting that maximises the information extracted from the observable. Consequently, both approaches converge to the same functional form.

This agreement validates both methods, indicating that the structure of the weighting reflects the underlying physics rather than an artefact of the optimisation procedure. Any residual differences can be attributed to the analytic derivation being performed in the continuous limit, whereas the numerical implementation is performed on a discrete grid.

If this framework is extended to include detector acceptance effects, particularly in cases where the acceptance is asymmetric, the analytic derivation would require modification to account for these effects. In such scenarios, the simplifications made in **Eqn. 9** would no longer hold, as the mean angle in the absence of an EDM would no longer vanish.

6.3 Implications for EDM Sensitivity

For the idealised dataset considered in this project, the analytic optimal weighting improves the sensitivity by 49.1%. This represents a significant gain, particularly given that EDM signals are expected to be extremely small and measurements are typically limited by statistical precision. The weighting therefore enables more efficient use of the available data, allowing additional information to be extracted from the same dataset without requiring further data collection.

More broadly, this approach is not restricted to this specific observable and can be applied to measurements where sensitivity varies across phase space. Weighting therefore provides a practical method for enhancing sensitivity in precision measurements, including EDM searches. The observed improvement demonstrates that a substantial fraction of the available sensitivity is encoded in the phase-space structure of the observable, suggesting that similar strategies may be applied in other precision experiments, such as collider measurements and dedicated EDM searches.

6.4 Limitations of the Analysis

The results presented in this analysis are obtained under idealised conditions. A key simplification is that the analytic PDFs were evaluated at Born-level, neglecting higher-order radiative corrections. Whilst this captures the leading behaviour, such effects may modify the precise form of the optimal weighting



in a more complete treatment.

Detector effects were also neglected, with the analysis assuming perfect angular and energy resolution. In practice, finite detector acceptance and resolution will alter the observed phase space distribution. Preliminary studies incorporating acceptance effects suggest that the overall structure of the weighting is preserved, with the dominant impact being a rescaling rather than a qualitative change. These effects are therefore not expected to alter the key qualitative features of the optimal weighting.

A further assumption in the generation of the PDFs is the use of the ultra-relativistic limit, in which the positron is treated as effectively massless with $v \approx c$. Whilst well justified for the energies considered, this approximation may introduce small deviations in a more complete treatment, although these are not expected to significantly affect the overall conclusions. In the absence of this approximation, the full kinematic range $\lambda \in [0.0, 1.0]$ could be considered, rather than restricting the analysis to $\lambda \in [0.2, 1.0]$.

From a numerical perspective, the optimisation was performed on discrete λ slices and combined to construct the full weight map. Together with the discretisation of the phase space into a 500×500 grid, this may introduce minor numerical artefacts. However, these are not expected to alter the overall structure of the optimal weighting.

Despite these limitations, the key features of the optimal weighting are expected to remain robust. The results therefore establish a clear proof of principle, with further refinement required for full experimental implementation. In particular, the inclusion of detector effects and higher-order corrections would provide a more realistic assessment of the achievable sensitivity in an experimental setting.

7 Conclusion

In this project, sensitivity to a fundamental particle's electric dipole moment (EDM) was improved through the introduction of an optimised weighting scheme, using a figure of merit to quantify performance. An optimal weighting function was derived that enhances the effective statistical power without requiring additional data, with direct application to the Fermilab $g-2$ experiment for extracting the rest-frame spin precession tilt δ .

The optimal weighting function was obtained numerically using the Covariance Matrix Adaptation Evolution Strategy (CMA-ES), and independently reproduced through an analytic derivation. The two approaches show close agreement, with the weight at $\lambda = 0.500$ exhibiting a cosine similarity of 0.996, indicating near-identical structure. The resulting weighting yields a 49.1% improvement in sensitivity relative to the unweighted case, corresponding to a substantial increase in effective statistical power. This structure reflects a balance between maximising sensitivity at large angles and retaining statistical precision from high event-rate regions.

The optimal weighting also enables the extraction of δ through a linear calibration between the gradient of the fitted amplitude and the EDM strength. Using this approach, the extracted value $\delta = (80.0 \pm 0.1)$ mrad is consistent with the injected value of 80 mrad. An independent log-likelihood analysis yields $\delta = (80.0 \pm 0.9)$ mrad, providing a consistent cross-check. The weighted method therefore recovers the correct value and improves the precision relative to the log-likelihood approach.

More broadly, this work demonstrates that a significant fraction of the available sensitivity is encoded in the phase-space structure of the observable. By exploiting this structure, the optimal weighting provides a physically motivated and effective route to enhancing sensitivity in precision EDM searches, with potential applicability to a wider class of measurements in which sensitivity varies across phase space.



8 Future Work

Future developments involve extending this framework to more realistic and experimentally applicable conditions.

A key extension is the inclusion of detector acceptance effects, particularly for non-symmetric or off-centre acceptance, where the simplifying assumptions used here no longer hold and the full figure of merit must be considered. This would allow the weighting scheme to be adapted to realistic detector geometries.

Further improvements include moving beyond the Born-level treatment by incorporating higher-order radiative corrections, providing a more complete description of the decay process.

The methodology can also be applied to simulated or real $g-2$ data, integrating the weighting framework into a full analysis pipeline. It may additionally be extended to dedicated muon EDM experiments, such as the proposed frozen-spin muEDM experiment [24], where the observable differs but the underlying optimisation problem remains the same.

From a numerical perspective, improvements include exploring alternative interpolation schemes or developing a fully continuous optimisation procedure, as well as alternative parameterisations of the weighting function.

Taken together, these extensions highlight the potential for this approach to be developed into a practical tool for future EDM measurements, enabling sensitivity to be assessed under realistic experimental conditions.



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9 Appendices

10 Code Repository

To support reproducibility, all code used in this project is available in the following GitHub repository:

`https://github.com/jake-dexter/mphys-masters-project`

The repository includes the full analysis workflow, covering data handling, weighting optimisation, observable extraction, and figure generation.

A Layperson's Description of the Project

The muon is a fundamental particle, meaning it is not made up of anything smaller. Because of this, it is not expected to have an electric dipole moment (EDM), which would correspond to a separation of electric charge across some distance. If a muon does have an EDM, it would appear as a very small asymmetry in how it decays, effectively producing a slight tilt in the distribution of its decay products.

Measuring this effect is extremely challenging, as the value predicted by the Standard Model is incredibly small ($\sim 10^{-38} \text{ e} \cdot \text{cm}$), and any signal is buried within a large amount of data. Importantly, not all of this data is equally useful: some regions contain far more information about a possible EDM than others.

The key idea behind this project is to make better use of the available data by giving more importance to the most informative regions. Rather than treating all data points equally, the analysis assigns greater weight to those that are most sensitive to the muon EDM.

By applying this approach, the sensitivity to the muon EDM is improved by approximately 50%, corresponding to a substantial increase in effective statistical power. This is equivalent to extracting more information from the same dataset, without requiring additional data collection.

This is important because observing a muon EDM would point to new physics beyond the Standard Model, and could help explain one of the biggest open questions in physics: why the Universe is dominated by matter rather than antimatter.

B Project Plan

Research Question:

What are the optimum positron decay angles and energies to measure in order to maximise the sensitivity of the muon EDM measurement at the Fermilab g-2 experiment?

Main aim and objective:

The primary aim of this project is to determine how to extract the maximum possible sensitivity to the muon electric dipole moment (EDM) from simulated decay data provided by my supervisor. Starting from the given probability density functions - which describe the positron decay angle and energy distributions - the goal is to derive an optimal weighting function that depends on both the decay angle (θ) and the positron energy (λ), obtained via manual entry, machine learning, and log-likelihood estimates. This weighting function will be designed to enhance the discrimination between events that contain a true EDM signal and those that do not. To define the weighting function appropriately we need to constrain all weights to be less than 1 and have it so that they all sum to 1.

To achieve this, I will first define and compute a figure of merit (FOM) - a quantitative measure that



reflects how effectively a particular weighting scheme distinguishes EDM-like events from background or null cases. The FOM will initially be evaluated using a range of trial weightings, and then optimised using machine learning techniques to identify the functional form that maximises the EDM sensitivity, with the current thought being to use unsupervised learning techniques such as neural networks. For each iteration of the ML-based optimisation, the corresponding FOM will be recalculated, allowing the training process to converge toward the most effective weighting function. A further step might be to calculate the FOM for the whole dataset, including each timestep, for a given weighting function.

In the later stages of the project, this approach will be extended by comparing the FOM-based method with one using a log-likelihood framework, applying the same optimisation principles to assess consistency and performance.

Through the use of simulated data and machine learning-driven optimisation, the project seeks to provide a data-driven strategy for improving the experimental reach of future muon EDM measurements.

I have created Gantt charts that reflect the timetable agreed upon by my supervisor and me, as outlined in full in my Week 4 log. The project naturally divides into two main phases: Semester 1 focuses on developing a deep understanding of the underlying physics and using machine learning techniques to optimise the figure of merit (FOM), whilst Semester 2 centres on applying a log-likelihood estimation to the data and comparing the outcomes to the results obtained in Semester 1.

My supervisor and I had a brief discussion regarding the mitigations of the project, and it was decided that some of the requirements of the project are not as essential as others. One example of this was that if the machine learning took too long to implement, it would be possible to drop the need for accounting for detector acceptance - which (θ, λ) pairs are relevant.

C Relativistic Muons

An example of a fundamental particle is the muon, μ^- , often referred to as a “heavy electron” due to being identical to the electron apart from its much larger mass, approximately 200 times greater [8]. Muons are typically produced in the upper atmosphere through cosmic ray interactions at altitudes of around 12 km above the Earth’s surface, travelling with velocities very close to the speed of light.

Classically, such a short mean lifetime of $\tau = 2.197 \mu\text{s}$ [25] would imply that muons decay long before reaching detectors at sea level. However, according to special relativity, the observed lifetime depends on the muon velocity relative to the Earth, $\beta = v/c$. In general, the spacetime coordinates in the laboratory frame, $(x^\mu) = (ct \ x \ y \ z)^T$, and in the muon rest-frame, $(x'^\mu) = (ct' \ x' \ y' \ z')^T$, are related by a Lorentz transformation:

$$x'^\mu = \Lambda^\mu{}_\nu x^\nu, \quad (24)$$

where $\gamma = (1 - \beta^2)^{-1/2}$ is the Lorentz factor with $\beta = |\boldsymbol{\beta}|$, and $\Lambda^\mu{}_\nu$ are the components of the Lorentz transformation. Without loss of generality, the coordinate system can be oriented such that the boost is along the x -axis, in which case the matrix representation of $\Lambda^\mu{}_\nu$ is given by:

$$(\Lambda^\mu{}_\nu) = \begin{pmatrix} \gamma & -\beta\gamma & 0 & 0 \\ -\beta\gamma & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \quad (25)$$

In this case, the Lorentz transformation leads to the familiar relations for time dilation and length contraction:



$$\tau_{\text{lab}} = \gamma \tau_{\text{rest}}, \quad L_{\text{lab}} = \frac{L_{\text{rest}}}{\gamma}. \quad (26)$$

Consequently, muons produced high in the atmosphere can be detected at ground level, as time dilation in the laboratory frame increases their apparent lifetime compared to classical expectations. In high-precision experiments such as Fermilab's $g-2$, the stored muons are likewise highly relativistic ($\gamma \approx 29.3$), and these effects are essential for accurately modelling their decay dynamics and spin precession during storage.

D Electromagnetic Field Tensor

In classical electromagnetism, Maxwell's equations relate the electric field $\mathbf{E}$, magnetic field $\mathbf{B}$, charge density ρ , and current density $\mathbf{j}$:

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}, \quad \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \quad \nabla \times \mathbf{B} = \mu_0 \left(\epsilon_0 \frac{\partial \mathbf{E}}{\partial t} + \mathbf{j} \right), \quad \nabla \cdot \mathbf{B} = 0. \quad (27)$$

These equations are not manifestly Lorentz covariant when written in terms of three-vectors. A covariant formulation is obtained by introducing the electromagnetic field tensor $F^{\mu\nu}$. Using the convention $(x^\mu) = (ct, x, y, z)$, the matrix representation of the components $F^{\mu\nu}$ is:

$$(F^{\mu\nu}) = \begin{pmatrix} 0 & -E_x/c & -E_y/c & -E_z/c \\ E_x/c & 0 & -B_z & B_y \\ E_y/c & B_z & 0 & -B_x \\ E_z/c & -B_y & B_x & 0 \end{pmatrix}. \quad (28)$$

In terms of $F^{\mu\nu}$, Maxwell's equations take the compact covariant form:

$$\partial_\mu F^{\mu\nu} = \mu_0 j^\nu, \quad \partial_{[\rho} F_{\mu\nu]} = 0, \quad (29)$$

where the four-current is defined by:

$$(j^\nu) = \begin{pmatrix} c\rho \\ \mathbf{j} \end{pmatrix}. \quad (30)$$

The tensor³ $F^{\mu\nu}$ transforms under Lorentz transformations as:

$$F'^{\mu\nu} = \Lambda^\mu{}_\alpha \Lambda^\nu{}_\beta F^{\alpha\beta}, \quad (31)$$

which ensures that Maxwell's equations retain the same form in all inertial frames.

For the case of relativistic muons in the Fermilab $g-2$ experiment, the laboratory-frame magnetic field can be written as $\mathbf{B} = (0, 0, B_z)^T$. Considering a momentarily co-moving rest frame of the muon, with instantaneous velocity $\mathbf{v} = (v_x, 0, 0)^T$, the electromagnetic fields transform according to the Lorentz transformation of the field tensor.

³Strictly speaking, the tensorial nature of $F^{\mu\nu}$ is not derived within classical electromagnetism, but postulated as part of its relativistic formulation. This assumption ensures that Maxwell's equations are Lorentz covariant.



In this frame, the fields become:

$$\mathbf{B}' = \begin{pmatrix} 0 \\ 0 \\ \gamma B_z \end{pmatrix}, \quad \mathbf{E}' = \begin{pmatrix} 0 \\ -\gamma\beta c B_z \\ 0 \end{pmatrix}. \quad (32)$$

Thus, a purely magnetic field in the laboratory frame induces a non-zero electric field in the muon rest frame. This is crucial for EDM searches, as a non-zero electric field can produce a torque on the muon in the presence of an electric dipole moment, as explained in the subsequent section **App. E**.

E Classical Dipole Moments

There are two types of dipole moments: electric, $\mathbf{d}$, and magnetic, $\boldsymbol{\mu}$. The electric dipole moment (EDM) is the more straightforward to visualise.

When two electric charges of equal magnitude but opposite sign, q , are separated by a displacement vector $\mathbf{r}$, the electric dipole moment is defined as:

$$\mathbf{d} = q \mathbf{r}. \quad (33)$$

Applying an external electric field $\mathbf{E}$ that is not aligned with $\mathbf{r}$ generates a torque $\boldsymbol{\tau}$, which acts to align the dipole with the field, given by:

$$\boldsymbol{\tau} = \mathbf{d} \times \mathbf{E}. \quad (34)$$

The magnetic dipole moment $\boldsymbol{\mu}$ of a current loop is defined as:

$$\boldsymbol{\mu} = I \mathbf{A} = \frac{q}{2m} \mathbf{L}, \quad (35)$$

where $\mathbf{A}$ is the area vector of the loop, and the second expression assumes a circulating charge q of mass m with angular momentum $\mathbf{L}$. When a magnetic field $\mathbf{B}$ is applied, the dipole experiences a torque that tends to align it with the field.

In both the electric and magnetic cases, the dipole experiences a torque that tends to align it with the applied external field, thereby minimising its potential energy:

$$U_E = -\mathbf{d} \cdot \mathbf{E}, \quad U_B = -\boldsymbol{\mu} \cdot \mathbf{B}. \quad (36)$$

Classically, dipole moments describe systems in which a spatial distribution of charge or current gives rise to an overall polarity. Electric dipoles arise from the separation of positive and negative charge, whilst magnetic dipoles originate from circulating currents or equivalent current loops. Both produce characteristic field patterns that fall off as $1/r^3$ and interact with external fields in an analogous manner.

This classical picture provides the foundation for understanding more complex systems, where similar concepts of electric and magnetic dipole moments can be extended to describe symmetry properties and, ultimately, intrinsic dipole moments at the particle level.

F Initial CMA-ES Test

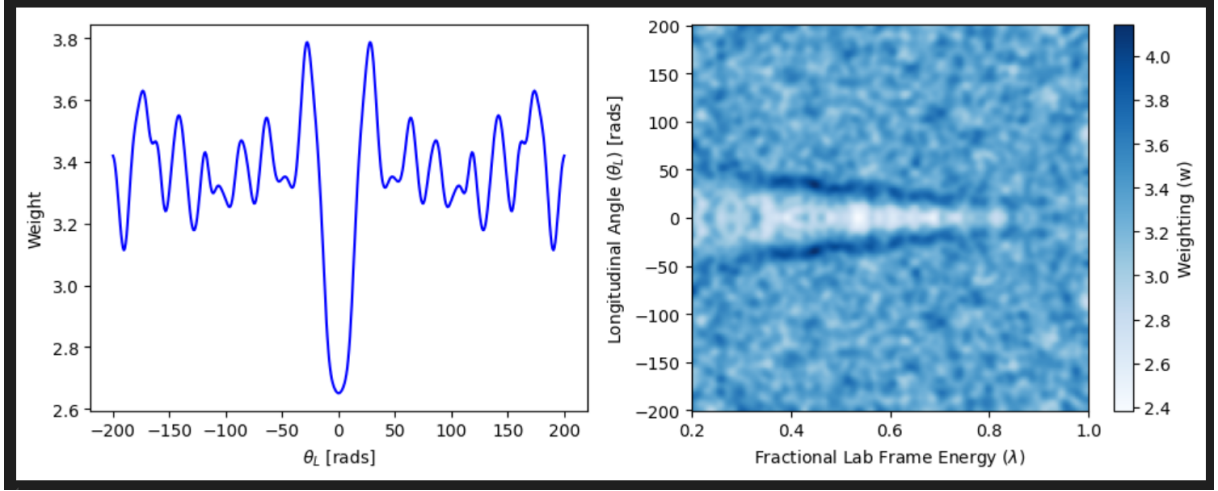


Figure 15: Initial CMA-ES optimisation using the full 500×500 grid. The kinematically allowed region shows the emergence of structure, whilst regions outside the kinematic maximum exhibit noise due to unconstrained parameters that do not contribute to the figure of merit.

An initial optimisation was performed using the full 500×500 grid for each λ slice. However, parameters corresponding to regions outside the kinematic maximum do not contribute to the figure of merit, as the probability density vanishes in these regions.

As a result, the optimiser assigns effectively unconstrained values to these parameters, leading to apparent random structure in the corresponding regions of the weight map. This obscures the physically meaningful behaviour within the kinematically allowed phase space. To address this, the optimisation was subsequently restricted to the region $|\theta_L| \leq \theta_{\max}(\lambda)$, with all weights outside this region fixed to zero. This reduces the dimensionality of the problem and ensures that the optimiser focuses only on physically relevant regions.

G Analytic Weight Derivation

To derive the analytic weight, the one-dimensional figure of merit is reformulated as a functional of the weight, as introduced in **Eqn. 15**. Throughout this derivation, the angular dependence is treated in the continuous limit, whilst the time index i remains discrete. In general, for a functional of the form:

$$F[w(\theta; \lambda)] = \int f(\theta, w(\theta; \lambda)) d\theta, \quad (37)$$

the functional derivative reduces to:

$$\frac{\delta F}{\delta w(\theta; \lambda)} = \frac{\partial f}{\partial w}(\theta, w(\theta; \lambda)), \quad (38)$$

provided that f depends locally on w and does not involve derivatives of w .



To construct the figure of merit, it is convenient to define the following intermediate functionals:

$$A_i[w; \lambda] = \int p_i(\theta; \lambda) w(\theta; \lambda) d\theta, \quad (39)$$

$$B_i[w; \lambda] = \int p_i(\theta; \lambda) w(\theta; \lambda) \theta d\theta, \quad (40)$$

$$C_i[w; \lambda] = \int p_i(\theta; \lambda) w(\theta; \lambda) \theta^2 d\theta, \quad (41)$$

$$D_i[w; \lambda] = \int p_i(\theta; \lambda) w(\theta; \lambda)^2 d\theta. \quad (42)$$

These allow the relevant observables to be written as:

$$y_i[w; \lambda] = \frac{B_i[w; \lambda]}{A_i[w; \lambda]}, \quad (43)$$

$$s_i^2[w; \lambda] = \frac{C_i[w; \lambda]}{A_i[w; \lambda]} - \left(\frac{B_i[w; \lambda]}{A_i[w; \lambda]} \right)^2, \quad (44)$$

$$n_{\text{eff},i}[w; \lambda] = \frac{A_i[w; \lambda]^2}{D_i[w; \lambda]}, \quad (45)$$

$$\sigma_i^2[w; \lambda] = \frac{s_i^2[w; \lambda]}{n_{\text{eff},i}} = \frac{(C_i[w; \lambda]A_i[w; \lambda] - B_i[w; \lambda]^2)D_i[w; \lambda]}{A_i[w; \lambda]^4}. \quad (46)$$

The figure of merit can then be expressed entirely in terms of these quantities. Taking the functional derivative with respect to $w(\theta; \lambda)$, and applying the chain, product, and quotient rules, yields the stationarity condition. After straightforward but lengthy algebra, this leads to the analytic weight given in **Eqn. 20**.

H Log-Likelihood Extraction of the EDM Parameter

An alternative approach to extracting the EDM parameter is to use a likelihood-based method. Since the data are provided as normalised probability densities rather than event counts with explicit per-bin uncertainties, the appropriate objective function is the multinomial log-likelihood (equivalently, the cross-entropy):

$$\log \mathcal{L}(\delta) = N \sum_i Z_i \log p_i(\delta) + \text{const}, \quad (47)$$

where Z_i denotes the measured, normalised probability density in bin i , satisfying $\sum_i Z_i = 1$, and $p_i(\delta)$ is the corresponding model prediction obtained from the analytic probability densities [21].

The factor N represents an effective event count, which sets the overall statistical scale of the likelihood without affecting the location of its maximum. In this analysis, it is fixed to $N = 10^6$ to ensure consistency between different extraction methods.

By scanning over values of δ , the likelihood is evaluated and the best-fit estimate $\hat{\delta}$ is obtained by maximising $\log \mathcal{L}$ (equivalently, minimising the negative log-likelihood). The uncertainty on $\hat{\delta}$ is determined using the standard likelihood-ratio criterion: for a single fitted parameter, the 1σ interval corresponds to values of δ for which the negative log-likelihood increases by one unit relative to its minimum. If the likelihood is approximately symmetric about the minimum, the uncertainty is taken as the average of the left and right crossings.

This likelihood-based extraction provides a direct comparison between the measured probability densities and the analytic model, allowing the EDM sensitivity to be quantified without introducing an ex-



plicit weighting scheme. Within this framework, the best-fit value of δ and its associated uncertainty can be extracted in a statistically well-defined manner.